

Group Presentations: Logic Programming with Prolog

Presentation schedule:

Presentation starts: March 31st - May 2nd

Group 1: Introduction to Logic Programming

Goal: Explain the logic programming paradigm and contrast it with imperative programming.

Presentation Content:

- Define programming paradigms and place logic programming within them.
- Explain declarative programming with examples.
- Introduce facts, rules, and queries.
- Walk through a simple family knowledge base.
- Code example:
 - `parent(john, mary).`
 - `parent(mary, anna).`
 - `grandparent(X, Y) :- parent(X, Z), parent(Z, Y).`
- In Class Activities:
 - Ask students to predict the result of a query before running it.
 - Run the query `grandparent(john, anna)` and discuss why it succeeds.
 - Have students write their own three facts and one rule about a domain such as friends or courses.
- Invite a few students to share their rules and test them live.

Group 2: Installing and Running Prolog

Goal: Guide students through installing and using SWI Prolog to run programs.

Presentation Content:

- Introduce Prolog and its role in logic programming.
- Explain why SWI Prolog is commonly used.
- Provide step by step installation instructions for different operating systems.
- Show how to open the interpreter and run it from terminal or command prompt.
- Explain how to create a .pl file and consult it.
- Example program and query execution.
- In Class Activities:
 - Students download and install SWI Prolog on their machines.
 - Students open the interpreter and confirm it runs correctly.

- Create a simple file named family.pl and load it.
- Run at least two queries and verify the outputs.
- Troubleshoot installation or execution issues with peers.

Group 3: Rules and Inference

Goal: Demonstrate how Prolog uses rules to perform logical inference.

- Presentation Content:
 - Explain the structure of rules using head and body.
 - Describe how Prolog evaluates rules.
 - Introduce the concept of inference.
 - Explain backtracking with a step by step example.
 - Provide sample rules and queries.
- In Class Activities:
 - Run a query with variables such as grandparent(X, anna).
 - Ask students to trace how Prolog finds the answer.
 - Modify a rule and observe how results change.
 - Have students create one new rule based on existing facts.
 - Test multiple queries and interpret results.

Group 4: Lists and Recursion

Goal: Teach how Prolog handles structured data and recursion.

Presentation Content:

- Explain list syntax and notation.
- Describe head and tail decomposition.
- Introduce recursion and how it replaces loops.
- Walk through a recursive example step by step.
- Example code for computing length of a list.
- In Class Activities:
 - Run the length predicate on different lists.
 - Ask students to manually trace recursive calls.
 - Have students write a rule to check if an element exists in a list.
 - Test the rule with multiple inputs.
 - Discuss common mistakes in recursion.

Group 5: Control and Search in Prolog

Goal: Explain how Prolog explores solutions using its search strategy.

Presentation Content:

- Describe Prolog execution model.
- Explain depth first search.
- Discuss how rule order affects results.
- Introduce the idea of multiple solutions.
- Provide examples where Prolog backtracks.
- In Class Activities:
- Run a query that produces multiple answers.
- Use the semicolon key to explore additional results.
- Change the order of rules and observe differences.
- Have students predict outputs before running queries.
- Discuss efficiency and possible infinite loops.

Group 6: Applications and Mini Project

Goal: Apply logic programming concepts to build a small knowledge based system.

Presentation Content:

- Discuss applications in artificial intelligence and expert systems.
- Explain how knowledge bases are structured.
- Present a simple diagnostic example.
- Show how rules combine to produce conclusions.
- In Class Activities:
- Run queries on the provided example and interpret results.
- Each group creates a small knowledge base with at least five facts.
- Define at least two rules that derive new information.
- Prepare three queries and test them live.
- Present the mini project and explain how the logic works.